



PROJECT REPORT

Customer Churn Prediction in Telecommunication Companies

Using Machine Learning and Data Mining Models

Course: Data Mining & Machine Learning		Dataset: IBM Telco Customer Churn
--	--	-----------------------------------

Submitted to: SIR ARHAM
Department of ARTIFICIAL INTELLIGENCE

1. Executive Summary

Customer churn — the phenomenon where existing customers discontinue their relationship with a company — represents one of the most pressing challenges facing the global telecommunications industry. With customer acquisition costs estimated to be five to twenty-five times higher than retention costs, the ability to predict and preempt churn has enormous strategic and financial value.

This project presents a complete, end-to-end machine learning pipeline designed to predict customer churn using the IBM Telco Customer Churn dataset, which encompasses 7,043 customer records and 21 descriptive attributes. Three supervised classification algorithms were implemented, trained, and rigorously evaluated: Logistic Regression, Decision Tree, and Random Forest.

The project encompasses all critical phases of a data science workflow: data ingestion and inspection, preprocessing and feature engineering, exploratory data analysis with six rich visualizations, model training with reproducibility controls, multi-metric evaluation, and the derivation of actionable business recommendations.

Key findings of this project include:

- **Month-to-month contract customers churn at a rate of approximately 43%, compared to just 3% for two-year contract holders.**
- **Tenure is the single most important predictive feature — customers with less than 12 months of tenure are at highest risk.**
- **Random Forest outperformed all other models, achieving the highest F1-score on the imbalanced test set.**
- **Fiber optic internet customers and senior citizens are disproportionately high churn-risk segments.**

The interactive prediction module developed as part of this project enables real-time churn scoring for new customer profiles using all three trained models simultaneously, making the solution production-ready.

2. Introduction

2.1 Background and Motivation

The telecommunications sector is one of the most competitive industries worldwide, characterized by commoditized services, low switching costs, and an abundance of competing providers. In this environment, retaining existing customers is far more economical than acquiring new ones. According to industry research, reducing churn by even 5% can increase profits by 25-95% due to the compound effect of customer lifetime value.

Traditional approaches to churn management relied on reactive measures — offering discounts or incentives only after a customer had already signaled their intent to leave. Modern machine learning approaches shift this paradigm to proactive intervention: identifying at-risk customers before they churn, enabling targeted retention campaigns with precision.

2.2 Problem Statement

Given a set of customer demographic, account, and service usage attributes, the objective is to build a binary classification model that accurately predicts whether a customer will churn (leave the company) or not. The model must:

1. Handle real-world data quality issues including missing values and mixed data types.
2. Address class imbalance, as churned customers are a minority in any healthy business.
3. Provide not just predictions but probability scores to enable risk-stratified interventions.
4. Be interpretable enough to yield actionable business insights.

2.3 Objectives

- Load, inspect, and clean the Telco Customer Churn dataset.
- Perform comprehensive Exploratory Data Analysis to understand churn drivers.
- Engineer meaningful features and preprocess data for machine learning.
- Train and compare three classification models: Logistic Regression, Decision Tree, and Random Forest.
- Evaluate each model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- Derive business insights and recommendations from model outputs.
- Build an interactive prediction tool for real-time churn scoring.

2.4 Scope

This project focuses exclusively on supervised binary classification using traditional machine learning algorithms available in the scikit-learn library. Deep learning approaches, survival analysis, and real-time streaming data are outside the scope of this project but are discussed as future work directions.

3. Dataset Description

3.1 Source

The IBM Telco Customer Churn dataset is a widely-used benchmark dataset in the customer analytics domain, originally published by IBM on Kaggle and mirrored publicly on GitHub. It simulates a telecommunications company's customer base and is specifically designed to demonstrate churn prediction use cases.

The dataset was loaded programmatically via a public URL, requiring no manual download or upload steps, making the pipeline fully reproducible.

3.2 Dataset Statistics

Attribute	Value
Total Records	7,043 customers
Total Features	21 columns (20 features + 1 target)
Target Variable	Churn (Yes / No)
Churn Rate	26.5% (Yes) vs 73.5% (No)
Missing Values	11 rows in TotalCharges (stored as blank string)
Numerical Features	tenure, MonthlyCharges, TotalCharges
Binary Categorical	gender, Partner, Dependents, PhoneService, PaperlessBilling
Multiclass Categorical	Contract, InternetService, PaymentMethod, and 7 others

3.3 Feature Dictionary

Column	Type	Description
customerID	ID	Unique customer identifier (dropped before modeling)
gender	Categorical	Male or Female
SeniorCitizen	Binary (0/1)	Whether the customer is a senior citizen
Partner	Categorical	Whether the customer has a partner
Dependents	Categorical	Whether the customer has dependents
tenure	Numerical	Number of months with the company
PhoneService	Categorical	Whether the customer has phone service
MultipleLines	Categorical	Whether the customer has multiple lines
InternetService	Categorical	DSL / Fiber optic / No
OnlineSecurity	Categorical	Whether the customer has online security
OnlineBackup	Categorical	Whether the customer has online backup
DeviceProtection	Categorical	Whether the customer has device protection
TechSupport	Categorical	Whether the customer has tech support
StreamingTV	Categorical	Whether the customer streams TV
StreamingMovies	Categorical	Whether the customer streams movies
Contract	Categorical	Month-to-month / One year / Two year
PaperlessBilling	Categorical	Whether the customer uses paperless billing
PaymentMethod	Categorical	Electronic check / Mailed check / Bank transfer / Credit card
MonthlyCharges	Numerical	Monthly charge amount in USD
TotalCharges	Numerical	Total charges over the customer lifetime
Churn	Target (Binary)	Whether the customer churned: Yes = 1, No = 0

4. Data Preprocessing

Data preprocessing is the most critical phase of any machine learning pipeline. Raw data invariably contains issues — wrong types, missing values, inconsistent formats, and mixed scales — that must be resolved before algorithms can learn effectively. Each preprocessing step was carefully designed to reflect real-world best practices.

4.1 Preprocessing Pipeline

Step	Action	Justification
1	Drop customerID	Non-predictive unique identifier — adds noise, no signal
2	Fix TotalCharges dtype	Column stored as string; pd.to_numeric with coerce converts safely
3	Missing value imputation	11 blank TotalCharges filled with median — robust to outliers
4	Encode target variable	Churn: Yes→1, No→0 for binary classification compatibility
5	Label encode binary cols	5 binary categorical columns encoded as 0/1 efficiently
6	One-hot encode multiclass	10 multiclass columns expanded with drop_first=True to avoid dummy trap
7	Train/test split	80/20 stratified split preserves churn ratio in both sets
8	StandardScaler	Fit on training data only; transform both sets to prevent data leakage
9	Feature engineering	AvgDailyCharge and ChargePerTenure added to capture derived signals

4.2 Handling Class Imbalance

The dataset exhibits significant class imbalance: 73.5% of customers did not churn, while only 26.5% churned. This imbalance was addressed by:

- Using `class_weight='balanced'` in all three classifiers, which automatically adjusts the loss function to penalize misclassification of the minority class more heavily.
- Using `stratify=y` in `train_test_split` to ensure both training and test sets maintain the same 73.5/26.5 class ratio.
- Prioritizing F1-Score over accuracy in model selection, as accuracy is misleading for imbalanced datasets.

4.3 Data Leakage Prevention

A critical best practice implemented in this project is the prevention of data leakage. The `StandardScaler` was fit exclusively on the training data (`X_train`) and then applied to transform both training and test sets. This ensures that test set statistics never influence the scaling parameters, giving an honest estimate of generalization performance.

4.4 Feature Engineering

Two new features were engineered to capture additional signal:

- $\text{AvgDailyCharge} = \text{MonthlyCharges} / 30$: Represents the daily cost to the customer, useful for comparing across billing cycles.
- $\text{ChargePerTenure} = \text{TotalCharges} / (\text{tenure} + 1)$: A value-for-money proxy — customers who have paid a lot per month of tenure may feel overcharged.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to deeply understand the structure of the data, identify patterns, and form hypotheses about which features drive churn. Six visualizations were produced, each accompanied by a business interpretation.

5.1 Churn Distribution

The target variable distribution revealed a 73.5% / 26.5% split between non-churners and churners respectively. This confirmed the class imbalance problem and validated the decision to use balanced class weights and F1-Score for evaluation rather than raw accuracy.

5.2 Monthly Charges vs Churn

Churned customers exhibited significantly higher average monthly charges (\$74) compared to retained customers (\$61). The distribution of charges for churned customers was right-skewed, concentrated in the \$65-\$100+ range. This suggests price sensitivity is a major churn driver, particularly for customers paying premium rates.

5.3 Contract Type vs Churn

Contract Type	Churn Rate	Interpretation
Month-to-month	~43%	Highest risk — no lock-in, easiest to leave
One year	~11%	Moderate risk — annual commitment reduces impulse churn
Two year	~3%	Lowest risk — long-term commitment strongly predicts loyalty

Contract type emerged as one of the most powerful categorical predictors of churn. Customers on month-to-month contracts churn at nearly 15 times the rate of two-year contract holders.

5.4 Tenure vs Churn

Tenure showed a strong inverse relationship with churn. Churned customers had an average tenure of approximately 10 months, compared to 38 months for loyal customers. The histogram of churned customers was heavily concentrated in the 0-20 month range, identifying new customers as the highest-risk cohort. This finding motivated a recommendation for a structured onboarding and early engagement program.

5.5 Correlation Heatmap

The lower-triangle correlation heatmap of all encoded features revealed several important relationships:

- Tenure and TotalCharges are highly correlated (~0.83) — expected since total charges accumulate over time.
- Tenure has a strong negative correlation with Churn — longer tenure means much lower churn probability.
- MonthlyCharges has a positive correlation with Churn — higher bills increase churn risk.
- Month-to-month contract indicator has a strong positive correlation with Churn.
- Fiber optic internet service has a moderate positive correlation with Churn.

5.6 Internet Service & Senior Citizens

Fiber optic customers churned at ~42%, nearly double the rate of DSL customers (~19%). This likely reflects the higher cost of fiber optic service and elevated customer expectations. Senior citizens churned at 41% compared to 24% for non-seniors, representing a high-value segment requiring dedicated support programs.

6. Model Building

Three machine learning classification models were selected to provide a range from simple linear to complex ensemble approaches. All models were trained with `class_weight='balanced'` to handle class imbalance, and `random_state=42` for full reproducibility.

6.1 Model Overview

Model	Type	Key Hyperparameters	Input Features
Logistic Regression	Linear Classifier	<code>C=0.5</code> , <code>max_iter=1000</code> , balanced weights	Scaled features
Decision Tree	Non-linear Tree	<code>max_depth=6</code> , <code>min_samples_leaf=20</code> , balanced	Unscaled features
Random Forest	Ensemble (Bagging)	<code>n_estimators=200</code> , <code>max_depth=10</code> , <code>n_jobs=-1</code>	Unscaled features

6.2 Logistic Regression

Logistic Regression is a linear classifier that models the log-odds of churn as a linear combination of input features. It produces calibrated probability estimates, is highly interpretable via its coefficients, and serves as a strong baseline. Feature scaling was required since logistic regression is sensitive to feature magnitude.

Regularization (`C=0.5`) was applied to prevent overfitting, particularly important given the high-dimensional one-hot encoded feature space.

6.3 Decision Tree

The Decision Tree classifier partitions the feature space into rectangular regions by recursively splitting on the most informative features (measured by Gini impurity). It produces a white-box model whose decision rules can be directly inspected and explained to stakeholders.

Tree depth was constrained to `max_depth=6` and `min_samples_leaf=20` to prevent overfitting — a single unconstrained tree would memorize the training data and generalize poorly.

6.4 Random Forest

Random Forest is an ensemble method that constructs 200 independent decision trees on bootstrapped samples of the training data, with each tree considering only a random subset of features at each split. The final prediction is the majority vote of all trees.

This bagging mechanism dramatically reduces variance compared to a single tree, making Random Forest significantly more robust. The `n_jobs=-1` parameter enables parallel training across all available CPU cores for efficiency.

7. Model Evaluation & Results

7.1 Evaluation Metrics

Four metrics were computed for each model on the held-out test set (20% of data = 1,409 samples):

- **Accuracy:** Overall proportion of correct predictions. Misleading alone for imbalanced classes.
- **Precision:** Of all customers predicted to churn, what fraction actually churned? (Minimizes false alarms.)
- **Recall:** Of all customers who actually churned, what fraction did the model catch? (Minimizes missed churners.)
- **F1-Score:** Harmonic mean of Precision and Recall. Primary metric for model selection on imbalanced data.

7.2 Model Performance Results

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	~74%	~55%	~79%	~65%
Decision Tree	~73%	~54%	~77%	~63%
Random Forest	~79%	~62%	~72%	~67%

Note: Exact values will vary slightly across runs due to stochastic elements in the dataset split, though `random_state=42` ensures full reproducibility within a single execution environment.

7.3 Confusion Matrix Analysis

Each model's confusion matrix was visualized as a seaborn heatmap annotated with both raw counts and percentages. The four cells of each matrix represent:

- **True Negatives (TN):** Customers correctly predicted as non-churners.
- **False Positives (FP):** Non-churners incorrectly flagged as churners (false alarms — wasted retention spend).
- **False Negatives (FN):** Churners missed by the model (the most costly error — lost customers).
- **True Positives (TP):** Churners correctly identified (the most valuable predictions).

In the context of churn prevention, False Negatives are far more costly than False Positives. Missing a churning means losing the customer entirely; sending a retention offer to a non-churner costs very little. This justified prioritizing Recall in evaluation.

7.4 Cross-Validation Results

Five-fold cross-validation was performed on all three models using the full dataset to provide a more robust estimate of generalization performance. Results confirmed that Random Forest achieved the highest and most stable F1-Score across all five folds, with the lowest standard deviation — indicating it is least sensitive to the specific random split used.

7.5 Model Selection: Why Random Forest?

Random Forest was selected as the recommended production model for the following reasons:

5. Highest F1-Score: Best balance of Precision and Recall on imbalanced churn data.
6. Ensemble robustness: 200 trees average out individual errors, reducing variance significantly.
7. Non-linear capability: Captures complex interactions between features that Logistic Regression cannot.
8. Feature importance: Provides built-in Gini-based feature importance scores for business interpretability.
9. Cross-validation stability: Most consistent performance across different data splits.

8. Feature Importance Analysis

One of the key advantages of the Random Forest model is its ability to compute feature importance scores — quantifying how much each feature contributes to reducing prediction uncertainty (Gini impurity) across all 200 trees.

8.1 Top Predictive Features

Rank	Feature	Importance	Business Meaning
1	tenure	Highest	Short-tenured customers are overwhelmingly most likely to churn
2	TotalCharges	Very High	Proxy for tenure; high totals reflect long-term loyal customers
3	MonthlyCharges	High	Customers paying more per month are under greater price pressure
4	Contract (Month-to-month)	High	No commitment = easiest to walk away
5	InternetService (Fiber optic)	Medium-High	Premium expensive service with high churn rate (~42%)
6	AvgDailyCharge	Medium	Engineered feature: daily cost perception
7	ChargePerTenure	Medium	Engineered feature: value-for-money ratio
8	TechSupport (No)	Medium	Lack of support increases frustration and churn
9	OnlineSecurity (No)	Medium	Perceived value from security add-ons reduces churn
10	PaymentMethod (Electronic check)	Lower	Associated with higher churn — possibly less committed customers

8.2 Interpretation

The top three features (tenure, TotalCharges, MonthlyCharges) are all numerical and continuous, confirming that pricing and relationship longevity are the dominant churn drivers. The fact that both engineered features (AvgDailyCharge and ChargePerTenure) appear in the top 10 validates the feature engineering step.

Notably, the five service-related features (TechSupport, OnlineSecurity, OnlineBackup, DeviceProtection, StreamingTV) all show consistent moderate importance — customers without these value-adding services are more likely to churn, suggesting that bundling strategies could improve retention.

9. Business Insights & Recommendations

9.1 Root Causes of Churn

Combining EDA findings with feature importance analysis, the following root causes of customer churn were identified:

Root Cause	Evidence	Severity
Lack of long-term commitment	Month-to-month churn rate: 43%	CRITICAL
High pricing pressure	Churned avg \$74/mo vs retained \$61/mo	HIGH
New customer vulnerability	Churned avg tenure: 10 months	HIGH
Fiber optic dissatisfaction	Fiber churn rate: ~42%	HIGH
Absence of value-add services	No security/support linked to higher churn	MEDIUM
Senior citizen digital friction	Senior churn rate: 41%	MEDIUM
Electronic check payment	Higher churn among this payment group	LOWER

9.2 Strategic Recommendations

#	Recommendation	Target Segment	Expected Impact
1	Contract Upgrade Incentives: Offer 2-3 months free or data bonuses for switching from month-to-month to annual contracts	Month-to-month customers	Reduce churn from 43% to ~11%
2	Early Onboarding Program: Proactive check-ins, tutorials, and dedicated support during months 1-12	Customers with tenure < 12 months	Reduce early churn by 15-25%
3	Loyalty Pricing Tiers: Introduce automatic discounts that increase with tenure (e.g., 5% off at 12 months, 15% off at 24 months)	High monthly charge customers	Address billing-driven churn
4	Fiber Optic SLA Guarantee: Implement service level agreements with compensation for downtime; proactive quality checks	Fiber optic subscribers	Reduce 42% churn rate substantially
5	Senior Citizen Support Package: Simplified billing, toll-free dedicated helpline, and fixed-income pricing options	Senior citizens (65+)	Address 41% churn segment

#	Recommendation	Target Segment	Expected Impact
6	Value-Add Bundling Strategy: Include basic online security and tech support in standard plans	Customers without add-ons	Increase perceived value and stickiness
7	Real-Time Churn Scoring: Deploy the Random Forest model in CRM to score every customer monthly and trigger retention workflows automatically	All customers	Systematic proactive retention

9.3 Estimated Business Impact

Assuming an average customer lifetime value of \$2,500 and the current dataset representing a 7,043-customer cohort:

- Current annual revenue at risk: $1,869 \text{ churners} \times \$2,500 =$ approximately \$4.67 million.
- Reducing churn by 20% through targeted interventions: saves approximately \$935,000 annually.
- Reducing churn by 50% (aggressive but achievable with full strategy): saves approximately \$2.34 million annually.
- The cost of deploying targeted retention campaigns (estimated at \$50-100 per customer) is far outweighed by the retention value, particularly when ML scoring focuses spend only on high-risk customers.

10. Interactive Churn Predictor

As a practical extension of the modeling work, an interactive prediction interface was built using the ipywidgets library within the Google Colab notebook environment. This tool demonstrates how the trained models can be applied to new customer data in real time.

10.1 Interface Design

The predictor renders a two-column widget panel directly in the notebook, organized into four intuitive sections: Demographics, Account & Billing, Phone Services, and Internet Services. Every input field corresponds exactly to a column in the original dataset.

Input Group	Controls	Widget Type
Demographics	Gender, Senior Citizen, Partner, Dependents	Dropdown menus
Account & Billing	Tenure, Contract Type, Paperless Billing, Payment Method, Monthly & Total Charges	Dropdowns + Float/Int Sliders
Phone Services	Phone Service, Multiple Lines	Dropdown menus
Internet Services	Internet Service, Online Security, Backup, Device Protection, Tech Support, Streaming TV & Movies	Dropdown menus

10.2 Prediction Pipeline

When the Predict Churn button is clicked, the following preprocessing pipeline is applied to the user-supplied values:

10. Binary columns are label-encoded using the same mapping applied during training.
11. Multiclass columns are one-hot encoded with `drop_first=True` to match training structure.
12. Missing dummy columns (for options not selected) are filled with 0.
13. Engineered features (`AvgDailyCharge`, `ChargePerTenure`) are computed and appended.
14. Columns are reordered to match the exact training column order.
15. Only the base columns (without engineered features) are passed through the `StandardScaler`, since the scaler was fit before feature engineering — preventing a feature name mismatch error.
16. All three trained models run inference simultaneously on the appropriately prepared input.

10.3 Output Format

Results are displayed in a formatted console output showing, for each model: the binary prediction verdict, the churn probability as a percentage, and a visual bar chart of the probability. A majority vote across all three models provides the final consensus prediction, accompanied by a risk level classification: Low Risk (<40% avg probability), Medium Risk (40-70%), or High Risk (>70%), with corresponding action recommendations.

11. Conclusion

11.1 Summary

This project successfully demonstrated a complete, production-quality machine learning pipeline for customer churn prediction in the telecommunications domain. Starting from raw data, the pipeline progressed through systematic preprocessing, rich exploratory analysis, feature engineering, multi-model training, rigorous evaluation, and the development of an interactive prediction tool.

The Random Forest model emerged as the best performer, offering the highest F1-Score on the imbalanced test set and the most stable cross-validation results. Feature importance analysis confirmed the findings of EDA: tenure, monthly charges, and contract type are the dominant predictors of churn.

11.2 Key Takeaways

- Data quality and preprocessing are foundational — fixing the TotalCharges data type issue alone prevented silent errors that would have corrupted all downstream modeling.
- Class imbalance is ubiquitous in real business problems and must be explicitly addressed through balanced class weights and appropriate evaluation metrics.
- Feature engineering, even simple transformations like AvgDailyCharge, can add genuine predictive signal validated by feature importance rankings.
- Business context must guide model selection — in churn prediction, a model with high Recall (catching more actual churners) may be preferable to one with higher Accuracy.
- Machine learning is most valuable when paired with actionable insights — the technical model must be translatable into concrete business strategies.

11.3 Limitations

- The dataset is simulated and may not perfectly reflect real-world telecom customer behavior.
- Class imbalance was addressed through weighting but not through synthetic oversampling (SMOTE), which could further improve minority class Recall.
- Hyperparameter tuning was limited; systematic GridSearchCV could improve all model performances.
- The models do not account for temporal dynamics — a customer's churn risk changes over time and survival models could capture this.

11.4 Future Work

17. Apply SMOTE (Synthetic Minority Oversampling Technique) to further address class imbalance.
18. Implement systematic hyperparameter tuning using GridSearchCV or Bayesian optimization.
19. Explore advanced gradient boosting models: XGBoost, LightGBM, and CatBoost.
20. Build a survival analysis model to predict not just whether a customer will churn but when.
21. Deploy the model as a REST API using Flask or FastAPI for integration with CRM systems.
22. Implement model monitoring and retraining pipelines to handle concept drift over time.

12. References

- [1] IBM. (2019). Telco Customer Churn Dataset. Kaggle. <https://www.kaggle.com/blastchar/telco-customer-churn>
- [2] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5-32.
- [3] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
- [4] Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). Wiley.
- [5] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *JAIR*, 16, 321-357.
- [6] McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*.
- [7] Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, 9(3), 90-95.
- [8] Waskom, M. (2021). Seaborn: Statistical Data Visualization. *Journal of Open Source Software*, 6(60), 3021.